Standard Operating Procedure for Addressing Routing Protocol Flap Failure (ALARM_4552) on Arista Access Nodes

Node: WFD-ACC-44 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_4552

Description

This Standard Operating Procedure outlines the steps for addressing the Routing Protocol Flap failure on the Arista Access node WFD-ACC-44, located at the Whitefield POP. The failure is attributed to the BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, all of which are indicators of a potential routing protocol instability. The Arista Access node, being a critical component of the network, is responsible for routing traffic between the core and edge networks. The BGP route churn occurs due to the dynamic nature of BGP, where route advertisements are exchanged between neighbors. OSPF adjacency loss can be caused by multiple factors, including network topology changes, interface flaps, or misconfigured OSPF parameters. ISIS LSP flooding, on the other hand, can be triggered by network topology changes, errors in the LSP database, or ISIS configuration issues. To address this issue, it is essential to identify the root cause of the problem and take corrective actions to stabilize the routing protocol and prevent further flapping.

Pre-requisite Checks

- check: Verify the power and cooling status at the Whitefield POP to ensure that the node is receiving adequate power and cooling.

- check: Check the redundant link status to ensure that there are no issues with the primary or secondary links.

- check: Verify the network topology to ensure that there are no changes or anomalies that could be causing the routing protocol flap.

Potential Root Causes

Cause	Probability
BGP route churn due to dynamic route advertisements.	High
OSPF adjacency loss due to network topology changes or interface flaps.	Medium
ISIS LSP flooding due to network topology changes or errors in the LSP database.	Low

Diagnostic Commands & Expected Outputs

```
Command: show bgp summary, Expected Output: BGP summary information, Description: This command displays a summary of the BGP neighbors, including the number of established and total connections. This information can help identify any issues with BGP neighbors or connections.
```

```
Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays information about the OSPF neighbors, including the state of the adjacency and the number of LSAs exchanged. This information can help identify any issues with OSPF neighbors or LSAs.

Command: show isis database, Expected Output: ISIS database information, Description: This command displays information about the ISIS database, including the number of LSPs and the state of the database. This information can help identify any issues with the ISIS database or LSPs.
```

Resolution Steps

Step 1: Step: Clear BGP session, Command: clear bgp *, Description: This command clears the BGP session, including the BGP neighbors and connections. This can help resolve issues related to BGP route churn.

Step 2: Step: Reset OSPF adjacency, Command: clear ospf neighbor *, Description: This command resets the OSPF adjacency, including the OSPF neighbors and LSAs. This can help resolve issues related to OSPF adjacency loss.

Step 3: Step: Clear ISIS LSP database, Command: clear isis database, Description: This command clears the ISIS LSP database, including the LSPs and errors. This can help resolve issues related to ISIS LSP flooding.

Post-Check Verification

```
Command: show bgp summary, Expected Output: BGP summary information, Description: This command verifies that the BGP session has been cleared and that the neighbors and connections are established.

Command: show ospf neighbor, Expected Output: OSPF neighbor information, Description: This command verifies that the OSPF adjacency has been reset and that the neighbors and LSAs are established.

Command: show isis database, Expected Output: ISIS database information, Description: This command verifies that the ISIS LSP database has been cleared and that the LSPs and errors are resolved.
```

Roll-back Procedures

- Step: Revert BGP configuration, Command: config replace startup-config, Description: This command reverts the BGP configuration to the previous state, including the BGP neighbors and connections.

- Step: Revert OSPF configuration, Command: config replace startup-config, Description: This command reverts the OSPF configuration to the previous state, including the OSPF neighbors and LSAs.

- Step: Revert ISIS configuration, Command: config replace startup-config, Description: This command reverts the ISIS configuration to the previous state, including the ISIS neighbors and LSPs.